

Local maximality of the HKO pentagon product

Let P be the regular pentagon of circumradius one, centred at the origin and with vertex $(1, 0)$. In coordinates (q_1, q_2, p_1, p_2) put $K_0 = P \times R(-\pi/2)P$, with symplectic form $\omega_0 = \sum_j dq_j \wedge dp_j$. The Haim–Kislev–Ostrover capacity formula for this body gives

$$c_{\text{EHZ}}(K_0) = 2 \cos(\pi/10)(1 + \cos(\pi/5)), \quad \text{sys}(K_0) = \frac{3 + \sqrt{5}}{5} > 1.$$

Here $\text{sys}(K) = c_{\text{EHZ}}(K)^2 / (2 \text{vol}_4 K)$. The counterexample itself is due to Haim–Kislev and Ostrover; the result proved here concerns its local maximality when the number of facets stays ten.

Theorem 1. *There is a Hausdorff neighborhood $\mathcal{U}$ of K_0 among convex four-dimensional polytopes with exactly ten facets such that $\text{sys}(K) \leq \text{sys}(K_0)$ for $K \in \mathcal{U}$. Equality holds precisely for those bodies in $\mathcal{U}$ obtained from K_0 by translation, positive dilation and a linear symplectic map.*

Touching upper bounds and the local maximum

The HKO body has systolic ratio greater than one. We now ask what happens when its ten supporting hyperplanes move. These perturbations include bodies that are no longer products: each of the ten normal vectors can move in all four coordinate directions. The result is local maximality among nearby polytopes with exactly ten facets, with equality precisely along the local family obtained by translations, positive dilations and linear symplectic maps. These transformations preserve the ratio, so they account for the equality cases.

Write the nearby bodies as

$$K(a) = \{x \in \mathbb{R}^4 : a_i \cdot x \leq 1, \quad i = 1, \dots, 10\}, \quad a = (a_1, \dots, a_{10}),$$

and let a_0 denote the HKO normals. We restrict a to a neighborhood in which the ten inequalities remain facets and the simple face structure is unchanged. At the end of the proof we justify why every sufficiently close ten-facet polytope has such a representation, using polarity to label its facets near those of HKO. The parameter space has dimension forty. Our proof constructs smooth functions $U_1, \dots, U_{26}$ satisfying

$$\text{sys}(K(a)) \leq U_j(a), \quad U_j(a_0) = \text{sys}(K(a_0)).$$

Thus each function touches the ratio from above at the HKO body. To prove that a nearby body has a smaller ratio, it suffices to find one of these upper bounds that has decreased. Different bounds may handle different directions of motion.

Here is the elementary mechanism. Suppose first that a function f on $\mathbb{R}^d$ has finitely many smooth upper bounds U_j that agree with it at the origin. Write $r_j = DU_j(0)$. Assume that these linear functionals span $(\mathbb{R}^d)^*$ and that

$$\sum_j \lambda_j r_j = 0, \quad \lambda_j > 0, \quad \sum_j \lambda_j = 1.$$

For every nonzero direction h , at least one $r_j(h)$ must be negative. Indeed, if all were nonnegative, their positively weighted sum could vanish only if they were all zero. The spanning assumption would then force $h = 0$.

This observation also gives a uniform decrease. The continuous function $h \mapsto \min_j r_j(h)$ is negative on the unit sphere. Compactness supplies $\eta > 0$ such that its value is at most $-\eta$

everywhere on that sphere. Since there are finitely many upper bounds, their differentiability gives a single neighborhood in which

$$|U_j(v) - U_j(0) - r_j(v)| \leq \frac{1}{2}\eta\|v\| \quad \text{for every } j.$$

For nonzero v in this neighborhood, choose a bound with $r_j(v) \leq -\eta\|v\|$. Then

$$f(v) \leq U_j(v) \leq f(0) - \frac{1}{2}\eta\|v\| < f(0).$$

The uniform estimate is what turns a statement about directions into a local maximum throughout a neighborhood.

At HKO, the derivatives cannot span all forty directions: the ratio is constant under its continuous symmetries. Their tangent space T is generated by four translation directions, one dilation direction and ten linear symplectic directions. The exact calculation checks that these fifteen vectors are independent; group dimension alone would not exclude a stabilizer at this body. The finite calculation shows that the twenty-six rows $DU_j(a_0)$ vanish on T , have rank twenty-five and admit the strictly positive relation above. Choose a linear complement S to T . Its dimension is twenty-five, and the restricted rows span S^* . The preceding argument proves a strict local maximum on the affine slice $a_0 + S$.

To recover all nearby parameters, apply a small symmetry to this slice. The derivative of this operation at a_0 combines the independent spaces T and S , so it is invertible. The inverse function theorem therefore expresses each nearby parameter as a symmetry of a nearby slice point. Invariance reduces its ratio to the ratio on the slice. It is smaller unless the slice point is a_0 , which gives the asserted equality cases.

It remains to construct the touching upper bounds and establish the finite rank and positivity statements. The variational capacity formula supplies the bounds: any feasible choice of weights with positive quadratic value Q gives $c_{\text{EHZ}} \leq 1/(2Q)$. We choose weights that attain the known capacity at a_0 , then hold selected weights fixed and solve the five linear equations for closure and normalization for the others. An invertible five-column minor makes this solution smooth, and strict positivity of the base weights makes positivity persist nearby. This constructs feasible weights even when the equations for a stationary optimum are singular. The explicit seven-facet example below shows how this continuation works and how to differentiate its upper bound.

For an ordered list $\sigma = (\sigma_1, \dots, \sigma_m)$ of distinct facets, write

$$C_\sigma(a) = \begin{pmatrix} a_{\sigma_1} & \cdots & a_{\sigma_m} \\ 1 & \cdots & 1 \end{pmatrix}, \quad e = (0, 0, 0, 0, 1)^T.$$

Its first four rows impose $\sum_i \beta_i a_{\sigma_i} = 0$ and its last row imposes $\sum_i \beta_i = 1$. Thus $C_\sigma \beta = e$ and $\beta_i > 0$ are precisely the closure, normalization and positivity conditions we use. Along any smooth feasible section with positive quadratic value, put

$$A_\sigma(a) = \frac{1}{2Q_\sigma(a, \beta(a))}, \quad U_\sigma(a) = \frac{A_\sigma(a)^2}{2V(a)}, \quad V(a) = \text{vol}_4 K(a).$$

The variational capacity inequality and positivity give $\text{sys}(K(a)) \leq U_\sigma(a)$. The exact calculations below establish equality at a_0 and determine the derivatives of these functions.

The concrete base point and its volume

Put $t = \tan(\pi/5)$, $s = (5 - t^2)/2 = \sqrt{5}$, and

$$t^4 - 10t^2 + 5 = 0, \quad 0 < t < 1, \quad \alpha = (3 - s)/2, \quad b = t(1 + s)/2, \quad d = s - 1.$$

The selected root is unique in $(0, 1)$: the polynomial is strictly decreasing there and changes sign. For the circumradius-one pentagon with vertex $(1, 0)$, the edge normals have arguments

$(2k+1)\pi/5$ and lengths $\sec(\pi/5) = d$. Rotating the second factor by $-\pi/2$ gives the following ten rows, in (q_1, q_2, p_1, p_2) order:

$$\begin{pmatrix} 1 & t & 0 & 0 \\ -\alpha & b & 0 & 0 \\ -d & 0 & 0 & 0 \\ -\alpha & -b & 0 & 0 \\ 1 & -t & 0 & 0 \\ 0 & 0 & t & -1 \\ 0 & 0 & b & \alpha \\ 0 & 0 & 0 & d \\ 0 & 0 & -b & \alpha \\ 0 & 0 & -t & -1 \end{pmatrix}.$$

The body is $K(a) = \{x : a_i \cdot x \leq 1\}$. Writing $\theta = \pi/5$, the area of one pentagon is the sum of five triangles with central angle 2θ , hence $B = \frac{5}{2} \sin(2\theta) = 5 \sin \theta \cos \theta$. Fubini therefore gives

$$V_0 = B^2 = 25 \sin^2 \theta \cos^2 \theta = \frac{25(5+s)}{32} = \frac{25(15-t^2)}{64}.$$

This product formula alone does not give the full-chart derivative. For an arbitrary perturbation $h_i \in \mathbb{R}^4$, differentiation of $(a_i + \epsilon h_i) \cdot (x + \epsilon u_i(x) a_i / \|a_i\|) = 1$ on facet F_i gives normal velocity $u_i(x) = -\langle h_i, x \rangle / \|a_i\|$. Integrating that velocity over the facets gives

$$DV_0[h] = - \sum_{i=1}^{10} \frac{\text{vol}_3(F_i)}{\|a_i\|} \langle h_i, \bar{x}_i \rangle.$$

Here F_i is an edge times the other centered pentagon. Its centroid is $\bar{x}_i = a_i / \|a_i\|^2$, its three-volume is $2 \sin \theta B$, and $\|a_i\| = \sec \theta$. Consequently

$$DV_0[h] = -\mu \sum_{i=1}^{10} a_i \cdot h_i, \quad \mu = 10 \sin^2 \theta \cos^4 \theta = \frac{25}{32} + \frac{5s}{16}.$$

This explicitly specifies all forty coefficients, including zero cross-factor coefficients, for arbitrary full-chart perturbations. Facet edges have measure zero in this first variation; local simple-polytope stability supplies the smooth volume chart.

One actual seven-facet feasible section

Use certificate entry 2 (the third entry of the zero-based witness), with

$$\sigma = (2, 3, 8, 9, 5, 4, 6), \quad I = (1, 2, 3, 4, 5), \quad J = (6, 7).$$

Indices in I, J are positions in the word, not facet labels. Set

$$r = \frac{6909830056250513}{10^{17}}, \quad c = \frac{t^2 + 5}{40}, \quad z = \frac{5 - t^2}{20} = \frac{s}{10}, \quad k = \frac{3 - t^2}{4}.$$

The exact fixed choices recovered by the existing verifier are $\beta_6 = r$, $\beta_7 = z$; the decimal-looking rational is an exact rational choice, not a rounding assertion. The complete base vector is

$$\beta_0 = (c + kr, c - r, c, c, z - kr, r, z)^T.$$

At this word the closure matrix's selected minor is

$$M = C_\sigma(a_0)_I = \begin{pmatrix} -\alpha & -d & 0 & 0 & 1 \\ b & 0 & 0 & 0 & -t \\ 0 & 0 & 0 & -b & 0 \\ 0 & 0 & d & \alpha & 0 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix}, \quad \det M = 10(1 - t^2) > 0.$$

The full matrix is obtained by adjoining columns $(a_4, 1)^T$ and $(a_6, 1)^T$. Writing $e = (0, 0, 0, 0, 1)^T$, define the nearby section by

$$\beta_J(a) = (r, z)^T, \quad \beta_I(a) = C_\sigma(a)_I^{-1}(e - C_\sigma(a)_J(r, z)^T).$$

Direct substitution gives $C_\sigma(a_0)\beta_0 = e$ and $\beta_0 > 0$. For example $0.726 < t < 0.727$ and $0.069 < r < 0.070$ suffice to check all seven signs from the displayed formulas. Invertibility and positivity persist in a neighborhood.

Use $\omega(x, y) = x_{q_1}y_{p_1} + x_{q_2}y_{p_2} - x_{p_1}y_{q_1} - x_{p_2}y_{q_2}$ and

$$Q_\sigma(a, \beta) = \sum_{j < i} \beta_i \beta_j \omega(a_{\sigma_j}, a_{\sigma_i}).$$

At the fixed base geometry, solving closure with arbitrary $(\beta_6, \beta_7) = (u, v)$ gives the particularly simple identity

$$Q_\sigma = -4tv^2 + (2t - 2t^3/5)v.$$

It is independent of u and has its maximum at $v = z$; its value is $Q_0 = t/5$. Thus its Hessian on the two-dimensional closure affine space is singular (in the (u, v) coordinates it is $\text{diag}(0, -8t)$). Nevertheless the five-by-five closure minor is invertible. No assertion that a nearby stationary or optimizing branch continues is needed: the displayed section remains feasible, and therefore supplies a smooth upper bound.

$$A_0 = \frac{1}{2Q_0} = \frac{5}{2t} = 5t - \frac{t^3}{2}, \quad U_0 = \frac{A_0^2}{2V_0} = \frac{11 - t^2}{10} > 1.$$

Identification $A_0 = c_{\text{EHZ}}(K(a_0))$ uses the existing HKO capacity formula $2 \cos(\pi/10)(1 + \cos(\pi/5))$, not the calculation of a feasible section by itself. Feasibility yields $c \leq A$, not the reverse bound.

A worked coordinate outside product-preserving perturbations

Take $h_2 = (0, 0, 1, 0)$ and $h_i = 0$ for $i \neq 2$: vary the first momentum coordinate of the second q -facet. This is flat coordinate 6 in the verifier's zero-based storage. Put $w = c + kr = \beta_{0,1}$. Differentiated closure gives

$$D\beta_J[h] = 0, \quad D\beta_I[h] = -wM^{-1}(0, 0, 1, 0, 0)^T = w \begin{pmatrix} t^3/20 - t/4 \\ t^3/20 - t/4 \\ t^3/40 - 3t/8 \\ -3t^3/20 + 5t/4 \\ t^3/40 - 3t/8 \end{pmatrix}.$$

Inserting these values into the product rule for Q gives the exact scalar

$$DQ[h] = \frac{3 - t^2}{80} + \left(\frac{3t^2}{20} - \frac{1}{4} \right) r - \frac{t^2 + 5}{4} r^2 =: q_h.$$

Since a_2 has zero momentum coordinates, $DV_0[h] = 0$, while

$$DA[h] = -\frac{25}{2t^2} q_h, \quad DU[h] = -\frac{125}{4t^3 V_0} q_h.$$

This is an actual coefficient of the upper-branch row, approximately -0.1801707325 . It illustrates why the forty-coordinate calculation is stronger than differentiating only within products.

A complete formula specification for all twenty-six rows

The accompanying table gives the exact fixed values for every section, including the worked entry. Except at entry 22, the minor positions are $I = (1, 2, 3, 4, 5)$ and $J = (6, \dots, m)$. Entry 22 instead has $I = (1, 2, 3, 4, 6)$ and $J = (5, 7)$. The table is ordered by the witness's zero-based certificate index. For each table row set $\beta_J = u$ and compute

$$\beta_I = C_I^{-1}(e - C_J u), \quad (D\beta)_J[h] = 0, \quad (D\beta)_I[h] = -C_I^{-1}(DC[h])\beta.$$

For each of the forty coordinate directions, compute

$$\begin{aligned} DQ[h] &= \sum_{j < i} ((D\beta_i[h])\beta_j + \beta_i(D\beta_j[h]))\omega(a_{\sigma_j}, a_{\sigma_i}) \\ &\quad + \sum_{j < i} \beta_i\beta_j(\omega(h_{\sigma_j}, a_{\sigma_i}) + \omega(a_{\sigma_j}, h_{\sigma_i})), \\ R[h] &= -\frac{A_0}{2V_0Q_0^2}DQ[h] - \frac{A_0^2}{2V_0^2}DV_0[h]. \end{aligned}$$

Together with the explicit table these formulas specify the entire 26×40 matrix R without floating hints, branch search, or printing 1040 expanded field entries.

The finite predicates are: every word has distinct valid labels; $\det C_I \neq 0$; $C\beta = e$; all $\beta_i > 0$; $Q = t/5$; the differentiated closure equations hold; the rows annihilate the fifteen symmetry columns; $\text{rank } R = 25$; and the one-dimensional kernel of R^T has a vector whose entries have one strict sign. Normalize that vector to sum to one, producing $\lambda > 0$, $\lambda^T R = 0$. The symmetry columns are explicitly $h_i = -a_{i,j}a_i$ for $j = 1, \dots, 4$, $h_i = -a_i$, and $h_i = -X^T a_i$ for the ten generators

$$X = \begin{pmatrix} A & B \\ C & -A^T \end{pmatrix}, \quad B = B^T, \quad C = C^T,$$

using the four matrix units for A and the three symmetric matrix units for each of B, C . Their column rank is required to be fifteen. These exact arithmetic predicates, followed by the positive-convex-margin and symmetry-slice arguments, prove the existing theorem. The geometric volume derivation and the capacity identification above are separate inputs to that implication; a program assigning their formulas does not prove them.

Why these coordinates cover the geometric neighborhood

We have proved the inequality in a neighborhood of the ordered list $a_0 = (a_{0,1}, \dots, a_{0,10})$. We finish by relating that neighborhood to the Hausdorff metric in the theorem. There are two points to check: nearby bodies admit nearby labelled normals, and these normals retain the face structure used to differentiate volume.

The origin stays in the interior under sufficiently small Hausdorff perturbations of $K_0 = K(a_0)$. One way to see this is through support functions: $h_K(u) = \max_{x \in K} u \cdot x$. Write B for the Euclidean unit ball. If $rB \subset K_0$ and the Hausdorff distance from K to K_0 is less than $\epsilon < r$, then $h_K(u) \geq h_{K_0}(u) - \epsilon \geq r - \epsilon$ for every unit vector u . Thus $(r - \epsilon)B \subset K$. On such a neighborhood polarity is continuous: the radial function, the distance from zero to the boundary along a unit direction, of K° is $1/h_K$ on the unit sphere, and the common positive lower bound on h_K makes these reciprocals converge uniformly. The polars are uniformly bounded convex bodies, so this also gives Hausdorff convergence.

The polar K_0° has the ten vertices $a_{0,i}$. Choose disjoint small balls about them. Each vertex is uniquely exposed by some linear functional ℓ_i . For polars sufficiently close to K_0° , every maximizer of ℓ_i lies in the corresponding ball. Otherwise a convergent sequence of maximizers outside that ball would limit to a maximizer on K_0° different from $a_{0,i}$. A maximizing face

contains a vertex. A polytope K with exactly ten facets has a polar with exactly ten vertices; hence there is exactly one polar vertex in each ball and none elsewhere. Labelling these vertices a_i gives

$$K = \{x : a_i \cdot x \leq 1, i = 1, \dots, 10\}, \quad a_i \longrightarrow a_{0,i} \quad \text{as } K \longrightarrow K_0.$$

For completeness, the face structure is stable in this neighbourhood. The base body is a product of two polygons and is simple: precisely four facets meet at each vertex, with independent normals. Their four equations therefore determine a smoothly varying intersection under a small change of a . All the remaining inequalities have strict slack at that intersection, so it remains a vertex with exactly those incident facets.

No new vertex can appear. The convex hull of the base normals contains a ball about zero; the convex hull of sufficiently close normals contains a smaller common ball. Consequently the bodies $K(a)$ are uniformly bounded. If arbitrarily close parameters had a new vertex, choose four independent active constraints there. There are finitely many four-element subsets, so along a subsequence the same subset is active and the vertices converge to a point of K_0 where those four constraints are equalities. Simplicity forces them to be the incident facets of one of the original vertices. Their intersection is the continuation already constructed, contradicting newness. This argument also excludes a new vertex arising from a subset whose normals are dependent at the base point; no invertibility assumption for such a subset is needed.

The vertex–facet incidences therefore remain unchanged. In particular each inequality still defines a facet, and a fixed triangulation of the boundary by these incidences expresses volume as a smooth function of the vertex coordinates. This justifies the smooth volume used above. Conversely, small changes of the ten normals give Hausdorff-small changes of $K(a)$: their convex hulls converge, contain a common ball about zero, and polarity is continuous there.

Finally, the parameter transformations used in the slice argument are explicit. Translation by y , dilation by $\rho > 0$, and a linear symplectic map M act, respectively, by

$$a_i \mapsto \frac{a_i}{1 + a_i \cdot y}, \quad a_i \mapsto \rho^{-1} a_i, \quad a_i \mapsto M^{-T} a_i.$$

They are smooth near the identity and have the fifteen derivative columns specified above. Their independence and the inverse function theorem give the local slice representation. The strict decrease on that slice now applies to every sufficiently close ten-facet polytope. Equality forces the slice parameter to vanish, and therefore forces the body to be a symmetry of K_0 . Conversely, every such symmetry preserves the ratio. This proves both assertions of the theorem.

The twenty-six exact choices

The constants $c = (t^2 + 5)/40$ and $z = (5 - t^2)/20$ are as above. Facet labels and positions start at one; entry numbers start at zero to identify the rows in the accompanying computational witness. Every finite decimal in the table denotes its exact rational value.

Entry	Word σ	Fixed positions J	Exact fixed values
0	(1, 2, 8, 4, 10, 6)	(6)	(c)
1	(1, 2, 8, 7, 4, 10)	(6)	(z)
2	(2, 3, 8, 9, 5, 4, 6)	(6, 7)	(0.06909830056250513, z)
3	(1, 8, 7, 3, 4, 10)	(6)	(z)
4	(2, 9, 8, 5, 4, 6)	(6)	(z)
5	(2, 8, 4, 5, 6, 10)	(6)	(c)
6	(1, 6, 2, 8, 5, 4, 10)	(6, 7)	(0.180901699437495, c)
7	(1, 7, 8, 4, 3, 9, 10)	(6, 7)	(0.0202254248593736825 t^2 + 0.0584220259843842225, 0.18090169943749473)
8	(1, 5, 7, 6, 3, 10, 9)	(6, 7)	(0.020225424859373655 t^2 + 0.058422025984384415, 0.18090169943749462)
9	(2, 8, 4, 3, 9, 5, 6)	(6, 7)	(0.1809016994374947, z)
10	(2, 3, 9, 8, 4, 5, 6)	(6, 7)	(0.18090169943749512, z)
11	(2, 8, 9, 5, 7, 6, 3)	(6, 7)	(0.18090169943749473, c)
12	(1, 7, 8, 4, 10, 2)	(6)	(c)
13	(1, 7, 4, 3, 9, 10)	(6)	(c)
14	(2, 9, 5, 6, 7, 3)	(6)	(c)
15	(2, 8, 3, 9, 5, 6)	(6)	(z)
16	(2, 8, 4, 5, 10, 6)	(6)	(c)
17	(1, 7, 3, 2, 8, 4, 10)	(6, 7)	(0.18090169943749507, z)
18	(1, 5, 6, 7, 3, 2, 9)	(6, 7)	(0.06909830056250547, z)
19	(1, 7, 3, 9, 5, 6, 2)	(6, 7)	(c , 0.06909830056250552)
20	(1, 8, 5, 4, 6, 10, 2)	(6, 7)	(c , 0.14926352961534042)
21	(1, 6, 2, 8, 7, 4, 10)	(6, 7)	(z , 0.1809016994374947)
22	(1, 5, 7, 4, 3, 9, 10)	(5, 7)	(0.18090169943749482, c)
23	(1, 7, 3, 4, 10, 9, 5)	(6, 7)	(c , 0.06909830056250518)
24	(1, 7, 6, 3, 10, 9, 5)	(6, 7)	(0.1809016994374948, c)
25	(2, 8, 9, 5, 4, 6, 3)	(6, 7)	(z , 0.12028993467659169)